TUYEN NGOC TRAN

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LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

Senior Computer Engineering student at UCF (graduating August 2026) with a unique combination of full-stack web development, embedded systems, PCB design, and robotics experience. Proficient in React, C, Java, and real-time systems (ESP32, MSP430, Jetson Orin Nano). Passionate about AI-driven software and bridging hardware-software integration. Brings a decade of professional client-facing experience, demonstrating exceptional communication, reliability, and team leadership.

EDUCATION

University of Central Florida — *Orlando, FL*

Bachelor of Science in Computer Engineering | Expected August 2026 | GPA: 3.1/4.0 | Dean's List (Spring 2024)

- Relevant Coursework: Embedded Systems, Object-Oriented Software Development, Computer Networking, Junior & Senior Design
- Minor: Vietnamese

Seminole State College — *Sanford, FL*

Associate in Arts — May 2021

PROJECT EXPERIENCE

Autonomous Feeding Robot Arm — UCF Senior Design I | 4-Member Team — *Spring 2026 – July 2026 (In Progress)*

- Collaborating in a 4-member team to build an autonomous feeding robot arm integrating myCobot 320, NVIDIA Jetson Orin Nano, and a USB camera for real-time object detection and precision motion control.
- Developed a companion web application for robot monitoring and control — designed UI/UX, integrated REST API endpoints to communicate with the robot controller.
- Implemented a computer vision pipeline using OpenCV for food item detection to guide arm positioning and grasping sequences.
- Tech stack: Python, React, OpenCV, myCobot 320 API, NVIDIA Jetson Orin Nano, REST APIs.
- Project website published; full system delivery on schedule for July 2026.

Custom PCB Design — LCD & Distance Sensor — UCF Junior Design — *Fall 2025*

- Designed a complete custom PCB from scratch using Fusion 360: captured schematic, routed board layout, verified component footprints, and assembled the physical prototype.
- Integrated an LCD display and ultrasonic distance sensor; validated real-time sensor readings displayed on-screen after assembly.
- Gained end-to-end hardware engineering experience: requirements → schematic → PCB layout → physical build → functional testing.

Ingredients Imposter — React App — Object-Oriented Software Development | 4- Member Team — *Summer 2025 – Present*

- Co-developed a React application with a MongoDB backend deployed on Google Cloud Platform.
- Built a real-time ingredient substitution engine using category and nutritional matching logic.
- Configured CI/CD pipeline via GitHub Actions to automate testing and streamline deployment.

Embedded Systems — MSP430 & ESP32 — *Aug 2024 – Dec 2024*

- Programmed and debugged real-time C applications on MSP430 microcontroller using Code Composer Studio; implemented timers, PWM, and interrupt-driven control loops.
- Leveraged ESP-WROOM-32 dual-core architecture for concurrent real-time tasks; integrated UART, I2C, and SPI communication protocols for peripheral interfacing.

TECHNICAL SKILLS

Languages: C, Java, JavaScript, HTML, CSS, Python

Frameworks & Tools: React, Node.js, MongoDB, REST APIs, Git/GitHub, GitHub Actions (CI/CD), Google Cloud Platform

Embedded & Hardware: ESP32, MSP430, NVIDIA Jetson Orin Nano, myCobot 320, PCB Design (Fusion 360), UART / I2C / SPI

Software & IDEs: VS Code, Code Composer Studio (CCS), Fusion 360, Multisim, Microsoft Office Suite

Concepts: Real-Time Systems, Computer Vision (OpenCV), Agile/Scrum, CI/CD, OOP

Languages (Spoken): English (fluent), Vietnamese (fluent)

WORK EXPERIENCE

Nail Technician — Self-Employed / Local Salon, Orlando, FL — *June 2013 – Nov 2023*

- Built and maintained a loyal clientele of 50+ customers through consistent quality, professionalism, and clear communication — habits directly transferable to client-facing and collaborative engineering roles.
- Managed scheduling, inventory, and day-to-day business operations in a fast-paced environment, developing strong time management and process optimization skills.
- Mentored and trained junior team members, demonstrating leadership ability and the capacity to communicate technical procedures clearly to others.